V spațiu vectorial

$v_1, \dots v_n \in V$

$\beta_1, \dots \beta_n \in \mathbb{R}$

$\beta_1 v_1 + \beta_2 v_2 + \dots + \beta_n v_n = 0$

$\Rightarrow \beta_1 = \beta_2 = \dots = \beta_n = 0$ unică soluție

$\Downarrow$ $\nexists$ rel. de dependență între vectori

$v_1, \dots v_n$ **sistem de generatori** pentru V $\to$ cai orice vector din V se poate scrie ca o combinație liniară a celorlalți vectori din V

Bază = sist. de gen + lin. ind.

Dimensiune = nr. vectori din bază

$\mathbb{R}^n \quad \dim \mathbb{R}^n = n$

$\mathbb{R}_{\le n}[x] \quad \dim \mathbb{R}_{\le n}[x] = n+1$

$\mathcal{M}_{m,n}(\mathbb{R}) \quad \dim \mathcal{M}_{m,n}(\mathbb{R}) = m \cdot n$

$$B_{\mathbb{R}^n} = \left\{ \begin{pmatrix}1\\0\\0\\\vdots\\0\\0\end{pmatrix}; \begin{pmatrix}0\\1\\0\\\vdots\\0\\0\end{pmatrix}; \dots \begin{pmatrix}0\\0\\\vdots\\1\\0\end{pmatrix}; \begin{pmatrix}0\\0\\\vdots\\0\\1\end{pmatrix} \right\}$$

$B_{\mathbb{R}_{\le n}[x]} = \{1; x; x^2 \dots x^n\}$

! Spațiu vectorial V
știm $\dim V = n$
n vect. $\in V$ liniar. ind. $\Rightarrow$ **formează bază**

① Verificați liniar independența vectorilor

a) în $\mathbb{R}^3$ $\quad v_1 = \begin{pmatrix}1\\2\\2\end{pmatrix} \quad v_2 = \begin{pmatrix}2\\1\\3\end{pmatrix} \quad v_3 = \begin{pmatrix}1\\2\\-6\end{pmatrix}$

Fie $\beta_1, \beta_2, \beta_3 \in \mathbb{R}$

$\beta_1 v_1 + \beta_2 v_2 + \beta_3 v_3 = 0_{\mathbb{R}^3}$

$$\beta_1 \begin{pmatrix}1\\2\\2\end{pmatrix} + \beta_2 \begin{pmatrix}2\\1\\3\end{pmatrix} + \beta_3 \begin{pmatrix}1\\2\\-6\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix} \Rightarrow \begin{cases} \beta_1 + 2\beta_2 + \beta_3 = 0 \\ 2\beta_1 + \beta_2 + 2\beta_3 = 0 \\ 2\beta_1 + 3\beta_2 - 6\beta_3 = 0 \end{cases}$$

$\text{rang}\begin{pmatrix}1&2&1\\2&1&2\\2&3&-6\end{pmatrix} = 3 \Rightarrow$ sistemul are sol unică $\Rightarrow$ (sist. omog.) $\Rightarrow \beta_1 = \beta_2 = \beta_3 = 0$

$-\cancel{6} + \cancel{8} + \cancel{6} - 2 - \cancel{6} + 24$

$\Rightarrow v_1, v_2, v_3$ lini. indep.

b) în $\mathbb{R}^3$ $\quad v_1 = \begin{pmatrix}1\\0\\1\end{pmatrix} \quad v_2 = \begin{pmatrix}3\\7\\5\end{pmatrix} \quad v_3 = \begin{pmatrix}-1\\-7\\-3\end{pmatrix}$

Var 1

$v_3 = -v_2 + 2v_1 \Rightarrow 2v_1 + (-1)\cdot v_2 + (-1)\cdot v_3 = 0_{\mathbb{R}^3}$

Verif. $2 + (-3) + (-1) = 0$
$0 + (-7) + 7 = 0$
$2 + (-5) + 3 = 0$

$\Rightarrow v_3$ depinde de v_1 și v_2

$\Rightarrow v_1, v_2, v_3$ lin. dep.

Var 2

Fie $\beta_1, \beta_2, \beta_3 \in \mathbb{R}$

$\beta_1 v_1 + \beta_2 v_2 + \beta_3 v_3 = 0_{\mathbb{R}^3}$

$$\beta_1 \begin{pmatrix}1\\0\\1\end{pmatrix} + \beta_2 \begin{pmatrix}3\\7\\5\end{pmatrix} + \beta_3 \begin{pmatrix}-1\\-7\\-3\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix} \Rightarrow \begin{cases} \beta_1 + 3\beta_2 - \beta_3 = 0 \\ 7\beta_2 - 7\beta_3 = 0 \\ \beta_1 + 5\beta_2 - 3\beta_3 = 0 \end{cases}$$

Fie $A = (v_1 | v_2 | v_3)$ matr. asociată

$\det A = \begin{vmatrix}1&3&-1\\0&7&-7\\1&5&-3\end{vmatrix} = -21 - 21 + 0 + 7 + 35 - 0 = 0$

$\Delta = \begin{vmatrix}1&3\\0&7\end{vmatrix} = 7 \neq 0$

$\Rightarrow \text{rang } A = 2$
$\Rightarrow$ Sist. comp. nedet.

$$\Rightarrow \begin{cases} \beta_1 + 3\beta_2 - \beta_3 = 0 \\ \beta_2 = \beta_3 \end{cases} \Rightarrow \begin{cases} \beta_1 + 2\beta_2 = 0 \\ \beta_2 = \beta_3 \end{cases} \Rightarrow \beta_2 = -\frac{\beta_1}{2} \Rightarrow \begin{cases} \beta_1 = -2\beta_2 \\ \beta_3 = \beta_2 \\ \beta_1 \in \mathbb{R} \end{cases}$$

$\Rightarrow v_1, v_2, v_3$ lin. dep.

$$\begin{pmatrix}\beta_1\\\beta_2\\\beta_3\end{pmatrix} = \begin{pmatrix}-2\beta_3\\\beta_3\\\beta_3\end{pmatrix} = \begin{pmatrix}-2\\1\\1\end{pmatrix}\beta_3, \quad \beta_3 \in \mathbb{R}$$

$\beta_3 = \frac{\sqrt{14}}{2} \Rightarrow \alpha_1 = \sqrt{14} \Rightarrow \alpha_2 = \frac{\sqrt{14}}{2} \Rightarrow -\sqrt{14}v_1 + \frac{\sqrt{14}}{2}v_2 + \frac{\sqrt{14}}{2}v_3 = 0$

c) $\mathbb{R}_{\le 2}[x]$ $\quad v_1 = x$
$v_2 = 2x - x^2$
$v_3 = 3x + 2x^2$

Fie $\beta_1, \beta_2, \beta_3 \in \mathbb{R}$

$\beta_1 v_1 + \beta_2 v_2 + \beta_3 v_3 = 0 \Rightarrow \beta_1 x + \beta_2(2x - x^2) + \beta_3(3x + 2x^2) = 0$

$$\Rightarrow \begin{cases} \beta_1 + 2\beta_2 + 3\beta_3 = 0 \\ -\beta_2 + 2\beta_3 = 0 \end{cases}$$

$\text{rang}\begin{pmatrix}0&0&0\\1&2&3\\0&-1&2\end{pmatrix} = 2 \Rightarrow$ vectori liniari depend.

d) $\mathcal{M}_2(\mathbb{R})$ $\quad v_1 = \begin{pmatrix}2&4\\-3&2\end{pmatrix} \quad v_2 = \begin{pmatrix}-1&-2\\3&1\end{pmatrix} \quad v_3 = \begin{pmatrix}1&2\\3&7\end{pmatrix}$

fie $\beta_1, \beta_2, \beta_3 \in \mathbb{R}$

$$\beta_1 v_1 + \beta_2 v_2 + \beta_3 v_3 = 0_2 \Rightarrow \begin{cases} 2\beta_1 - \beta_2 + \beta_3 = 0 \\ 4\beta_1 - 2\beta_2 + 2\beta_3 = 0 \\ -3\beta_1 + 3\beta_2 + 3\beta_3 = 0 \\ 2\beta_1 + \beta_2 + 7\beta_3 = 0 \end{cases}$$

$\begin{vmatrix}2&-1&1\\-3&3&3\\2&1&7\end{vmatrix} = 42 - 6 - 3 - 6 - 6 - 21 = 21 - 21 = 0 \Rightarrow$

$\Delta = \begin{vmatrix}2&-1\\-3&3\end{vmatrix} = 6 + 3 = 9 \neq 0$

$\Rightarrow \text{rang } A = 2 \Rightarrow$

$\Rightarrow \text{rang} < 3 \Rightarrow v_1, v_2, v_3$ lin. dep.

e) V = sp. fct cont pe $\mathbb{R}$

$v_1 = 1 \quad v_2 = \cos t \quad v_3 = \cos^2 t \quad v_4 = \cos^3 t$

fie $\beta_1, \beta_2, \beta_3, \beta_4 \in \mathbb{R}$

$\beta_1 v_1 + \beta_2 v_2 + \beta_3 v_3 + \beta_4 v_4 = 0 \quad (\forall)\, t \in \mathbb{R}$

fie $t = 0 \Rightarrow \beta_1 \cdot 1 + \beta_2 \cos 0 + \beta_3 \cdot \cos^2 0 + \beta_4 \cdot \cos^3 0 = 0$

$t = \frac{\pi}{4} \Rightarrow \beta_1 \cdot 1 + \beta_2 \cdot \cos\frac{\pi}{4} + \beta_3 \cdot \cos^2\frac{\pi}{4} + \beta_4 \cos^3\frac{\pi}{4} = 0 \Rightarrow$

$t = \frac{\pi}{2} \Rightarrow \beta_1 \cdot 1 + \beta_2 \cos\frac{\pi}{2} + \beta_3 \cdot \cos^2\frac{\pi}{2} + \beta_4 \cos^3\frac{\pi}{2} = 0$

$t = \pi \Rightarrow \beta_1 \cdot 1 + \beta_2 \cos\pi + \beta_3 \cdot \cos^2\pi + \beta_4 \cdot \cos^3\pi = 0$

$$\Rightarrow \begin{cases} \beta_1 + \beta_2 + \beta_3 + \beta_4 = 0 \\ \beta_1 + \frac{\sqrt{2}}{2}\beta_2 + \left(\frac{\sqrt{2}}{2}\right)^2\beta_3 + \left(\frac{\sqrt{2}}{2}\right)^3\beta_4 = 0 \\ \beta_1 = 0 \\ \beta_1 - \beta_2 + \beta_3 - \beta_4 = 0 \end{cases} \Rightarrow \begin{cases} \beta_1 = 0 \\ \beta_2 + \beta_3 + \beta_4 = 0 \\ \beta_2 + \frac{\sqrt{2}}{2}\beta_3 + \left(\frac{\sqrt{2}}{2}\right)^3\beta_4 = 0 \\ -\beta_2 + \beta_3 - \beta_4 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} \beta_1 = 0 \\ L_2 + L_4 \Rightarrow \beta_3 = 0 \\ \beta_2 + \beta_4 = 0 \\ \beta_2 + \left(\frac{\sqrt{2}}{2}\right)^3\beta_4 = 0 \end{cases} \Rightarrow \left[\left(\frac{\sqrt{2}}{2}\right)^3 - 1\right]\beta_4 = 0 \Rightarrow \beta_4 = 0 \Rightarrow \beta_2 = 0$$

$\Rightarrow \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0 \Rightarrow v_1, v_2, v_3, v_4$ vect. lin. ind.

② În $\mathbb{R}^3$ verificați dacă $v_1 = \begin{pmatrix}1\\1\\1\end{pmatrix} \quad v_2 = \begin{pmatrix}0\\1\\1\end{pmatrix} \quad v_3 = \begin{pmatrix}0\\0\\1\end{pmatrix}$

$\mathbb{R}^3 = Sp\{v_1, v_2, v_3\} \Leftrightarrow (\exists)\, \beta_1, \beta_2, \beta_3 \in \mathbb{R}$ a.î. $(\forall)\, v \in \mathbb{R}^3$
$\Rightarrow v = \beta_1 v_1 + \beta_2 v_2 + \beta_3 v_3$

Fie $v = \begin{pmatrix}x\\y\\z\end{pmatrix} \in \mathbb{R}^3$

$$\beta_1 \begin{pmatrix}1\\1\\1\end{pmatrix} + \beta_2 \begin{pmatrix}0\\1\\1\end{pmatrix} + \beta_3 \begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}x\\y\\z\end{pmatrix} \Rightarrow \begin{cases} \beta_1 = x \\ \beta_1 + \beta_2 = y \\ \beta_1 + \beta_2 + \beta_3 = z \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} \beta_1 = x \\ \beta_2 = y - x \\ \beta_3 = z - y \end{cases} \Rightarrow (\exists)\, \beta_1, \beta_2, \beta_3 \in \mathbb{R} \text{ a.î. } (\forall)\, v \in \mathbb{R}^3 \Rightarrow v = \beta_1 v_1 + \beta_2 v_2 + \beta_3 v_3$$

$\Rightarrow v_1, v_2, v_3$ sist. de gen. pt. $\mathbb{R}^3$

③ Verificați dacă B bază în V. În caz contrar extrageți o bază sau completați până la o bază

a) $\mathbb{R}_{\le 2}[x] \quad \dim_{\mathbb{R}} V = 3$

$B = \{\underbrace{1+x}_{v_1}, \underbrace{2 - x + x^2}_{v_2}, \underbrace{3x - 2x^2}_{v_3}, \underbrace{-1 + 3x + x^2}_{v_4}\}$

B are 4 vectori
$\dim_{\mathbb{R}} V = 3 < 4$
$\Rightarrow$ B nu e bază, ci mai mult decât o bază $\Rightarrow$ extragem o bază din ea

$$\begin{pmatrix}1&2&0&-1\\1&-1&3&3\\0&1&-2&1\end{pmatrix} \xrightarrow{L_2 - L_1} \begin{pmatrix}1&2&0&-1\\0&-3&3&4\\0&1&-2&1\end{pmatrix} \xrightarrow{L_3 + \frac{1}{3}L_2}$$

$$\to \begin{pmatrix}1&2&0&-1\\0&-3&3&4\\0&0&-1&\frac{7}{3}\end{pmatrix}$$

$v_1 \quad v_2 \quad v_3 \Rightarrow$ formează o bază pt. că sunt vect. liniari

b) $\mathbb{R}^3 = V \quad B = \left\{\begin{pmatrix}1\\1\\0\end{pmatrix}; \begin{pmatrix}2\\0\\1\end{pmatrix}\right\} \quad \dim_{\mathbb{R}} V = 3$

$B_{\mathbb{R}^3} = \left\{\begin{pmatrix}1\\0\\0\end{pmatrix}; \begin{pmatrix}0\\1\\0\end{pmatrix}; \begin{pmatrix}0\\0\\1\end{pmatrix}\right\} \quad$ Fie $v = \begin{pmatrix}1\\0\\0\end{pmatrix}$

Verif. dacă v_1, v_2, v liniari indep $\Rightarrow \begin{vmatrix}1&2&1\\1&0&0\\0&1&0\end{vmatrix} = 1 \neq 0$

$\Rightarrow v_1, v_2, v$ sunt liniari indep. și formează o bază

c) $\mathbb{R}_{\le 2}[x] = V \quad \dim_{\mathbb{R}} V = 3$

$B = \{\underbrace{1+x}_{v_1}, \underbrace{1-x}_{v_2}\} \qquad B_V = \{1, x, x^2\}$

! lipsește x^2 care e obligatoriu $\to$ pe el îl vom lua
la fel și la $\begin{pmatrix}1\\0\\1\end{pmatrix}; \begin{pmatrix}7\\0\\5\end{pmatrix}$ vom adăuga $\begin{pmatrix}0\\1\\0\end{pmatrix}$

Fie $v = x^2$

Verific. liniar indep. v_1, v_2, v. $\quad \begin{vmatrix}1&1&0\\1&-1&0\\0&0&1\end{vmatrix} = -1 - 1 = -2 \neq 0$

$\Rightarrow v_1, v_2, v$ liniar ind $\Rightarrow \{v_1, v_2, v\}$ bază

④ Determinați $Ker(A)$ și $Im(A)$ (bază + dimensiune)

pt. $A = \begin{pmatrix}1&0&1\\2&2&3\\0&-2&-1\end{pmatrix}$

a) $Ker(A) = \{v \in \mathbb{R}^3 \mid A \cdot v = 0_{\mathbb{R}^3}\}$

$$\begin{cases} x + z = 0 \\ 2x + 2y + 3z = 0 \\ -2y - z = 0 \end{cases} \qquad \det A = -2 - 4 + 6 = 0, \quad \Delta_c = \begin{vmatrix}1&0\\2&2\end{vmatrix} = 2 \neq 0 \Rightarrow \text{S.C.N.}$$

Notăm $z = \beta$ (nec. sec.) $\beta \in \mathbb{R}$ $\Rightarrow \begin{cases} x = -\beta \\ 2x + 2y = -3\beta \\ -2y = \beta \Rightarrow y = -\frac{\beta}{2} \end{cases}$

$$\Rightarrow Ker(A) = \left\{\begin{pmatrix}-\beta\\-\frac{\beta}{2}\\\beta\end{pmatrix} \middle| \beta \in \mathbb{R}\right\} = Sp\left\{\begin{pmatrix}-2\\-1\\2\end{pmatrix}\right\}$$

$\dim_{\mathbb{R}} Ker(A) = 1$

b) $Im(A) = \{w \in \mathbb{R}^3 \mid (\exists)\, v \in \mathbb{R}^3 \text{ a.î. } Av = w\}$

$Im(A) = \{\underbrace{\begin{pmatrix}1\\2\\0\end{pmatrix}}_{v_1}; \underbrace{\begin{pmatrix}0\\2\\-2\end{pmatrix}}_{v_2}; \underbrace{\begin{pmatrix}1\\3\\-1\end{pmatrix}}_{v_3}\}$

$\begin{vmatrix}1&0&1\\2&2&3\\0&-2&-1\end{vmatrix} = -2 - 4 + 6 = 0 \Rightarrow$ sunt liniari dependenți
$\Rightarrow$ se va extrage o bază

Verific v_1, v_2 liniari independenți
v_1 și v_2 nu sunt proporționali
$\Rightarrow$ adevărat $\Rightarrow$

$\Rightarrow v_1$ și v_2 formează bază în spațiul $Im(A)$

$\dim_{\mathbb{R}} Im(A) = 2$